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Lightweight Improved YOLO26 for UAV-Based Face Detection and Recognition

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***Abstract*—In practical UAV scenarios, tiny faces (10–30 pixels), motion blur, and complex backgrounds severely degrade face detection and recognition performance, limiting its deployment in public security and other fields. To address this, we propose a Lightweight Improved YOLO26 detector, consisting of an eight-layer multi-scale feature extraction network, an enhanced feature fusion unit (EFFU), and a lightweight CNN face recognition branch. Experiments on the WIDER FACE dataset show that the model outperforms YOLO11-N and original YOLO26 in detection and recognition rates, while maintaining lightweight and real-time performance, making it suitable for UAV airborne deployment.**

***Index Terms*—UAV; face detection; face recognition; lightweight neural network; multiscale feature fusion**

I. INTRODUCTION

In recent years, UAV-based visual perception technology has been widely applied in public security, emergency rescue, border patrol, and intelligent inspection due to its advantages of flexible movement and wide monitoring range [1]. However, face targets captured by UAVs are mostly tiny-scale (10–30 pixels), accompanied by motion blur and complex background interference, which severely degrade the performance of existing detection and recognition models [2]. This problem has become a key bottleneck restricting the practical application of UAV face detection technology.

Current mainstream methods have significant limitations: two-stage image enhancement-based methods suffer from error accumulation and high computational complexity[3]; the state-of-the-art YOLO26[4] lacks sufficient tiny-target feature extraction capability and has suboptimal lightweight performance[5]; most detectors lack dedicated face recognition branches, making end-to-end inference difficult[6].

The main contributions of this paper are summarized as follows:

- Improve the backbone feature extraction network of YOLO26 by adopting depthwise separable convolution as the basic unit, which enhances the fine-grained feature perception for tiny faces and improves the detection rate while reducing model parameters.
- Propose an enhanced feature fusion unit (EFFU) to effectively integrate shallow detail features and deep semantic features, suppress background noise interference, and solve the problem of scale mismatch in multi-scale feature fusion.
- Construct a lightweight end-to-end face recognition network, which is tightly coupled with the detection module to achieve integrated detection and recognition with high recognition accuracy and low computational cost.
- Maintain low model parameters and high inference speed, ensuring that the model can be stably deployed on UAV airborne platforms with limited computing resources.

II. RELATED WORK

A. UAV-Based Face Detection

UAV face detection is constrained by complex imaging environments[7] and limited airborne computing resources. Tiny targets, motion blur, and background interference are core challenges[8]. Image preprocessing methods improve quality but increase latency, while attention mechanisms and multi-scale feature fusion have become mainstream optimization directions[9].

B. YOLO Series Detectors

YOLO series detectors are classical one-stage real-time detection frameworks, which are widely used in various visual detection tasks due to their high efficiency[10]. YOLO11-N and YOLO26, as the latest versions, have made certain progress in inference speed and detection accuracy, but their feature extraction ability for tiny targets is still insufficient, and their lightweight degree needs further improvement to meet the deployment requirements of UAV airborne platforms[11].

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C. Multi-Scale Feature Fusion

Multi-scale feature fusion technology integrates shallow detail information and deep semantic information to make up for the deficiency of single-scale features in target detection[12]. Traditional fusion structures (such as concatenation and addition) have low information utilization efficiency, while attention-based fusion modules can adaptively weight effective features, effectively enhancing target features and improving the detection rate of tiny targets [8].

D. End-to-End Face Recognition

Traditional separate detection-recognition frameworks suffer from error transmission and low efficiency. End-to-end frameworks realize synchronous detection and identity classification, simplifying inference and reducing errors, which is more suitable for UAV real-time requirements[11].

III. PROPOSED METHOD

The proposed lightweight improved YOLO26 adopts a three-stage progressive inference framework comprising image preprocessing, face detection, and face recognition. This design achieves standardized input processing, high-precision tiny face localization, and efficient identity classification, meeting the lightweight and real-time requirements of UAV airborne platforms. The overall flowchart of the model is shown in Fig. 1.

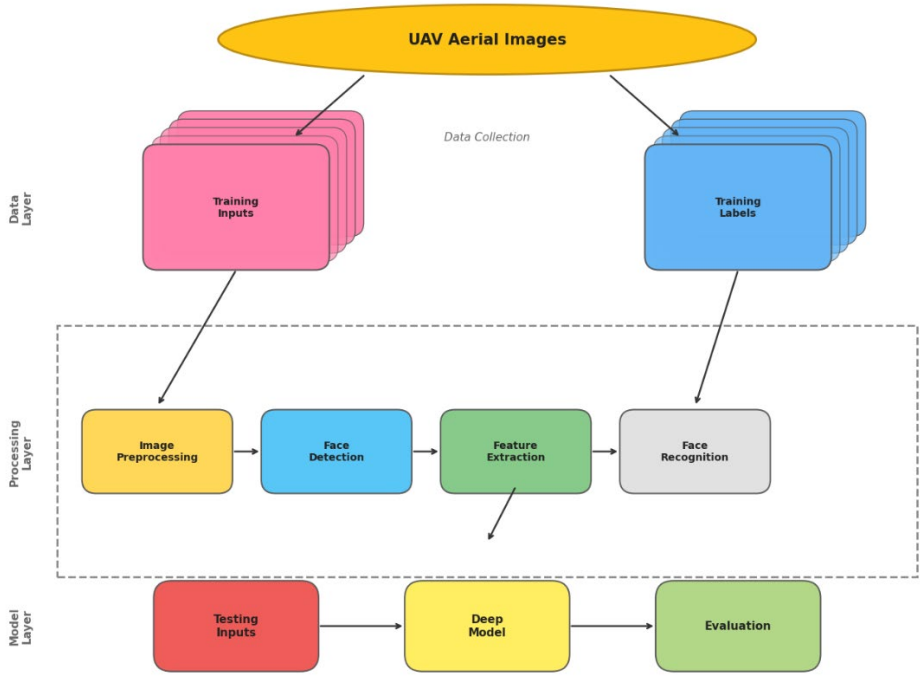


Fig.1 Overall flowchart

A. Image Preprocessing

Image preprocessing is used to eliminate the differences of input images caused by UAV imaging equipment, shooting distance, and illumination changes, laying a foundation for subsequent feature extraction and recognition. It includes three standardized operations:

1) Size Normalization

Original UAV aerial images are uniformly scaled to 640 x 640 pixels to adapt to the network input size and ensure the consistency of feature perception across different images.

2) Pixel Normalization

Pixel values [0, 255] are mapped to [0, 1] by linear transformation, which can accelerate the gradient descent process and improve the model convergence speed.

3) Data Augmentation

During the training process, random horizontal flip, color gamut jitter, and scale scaling are adopted to expand the diversity of training data, inhibit model over-fitting, and enhance the robustness of the model in complex UAV imaging scenarios.

B. Face Detection

Optimized based on YOLO26, the detection module consists of a lightweight backbone, 8-layer multi-scale feature extraction, EFFU, and detection head.

1) Network Architecture

The improved hierarchical face detection network comprises a lightweight backbone, 8-layer multi-scale feature extractor, EFFU, and detection head, balancing efficiency and tiny face detection performance The specific architecture of the improved face detection network is shown in Fig.2.

2) Lightweight Backbone Network

Replace CSPDarknet53 with DSConv-based backbone, which decomposes standard convolution into depthwise and pointwise convolution to reduce parameters, Its mathematical expression is:

$$DSConv(X) = PWConv(DWConv(X)) \quad (1)$$

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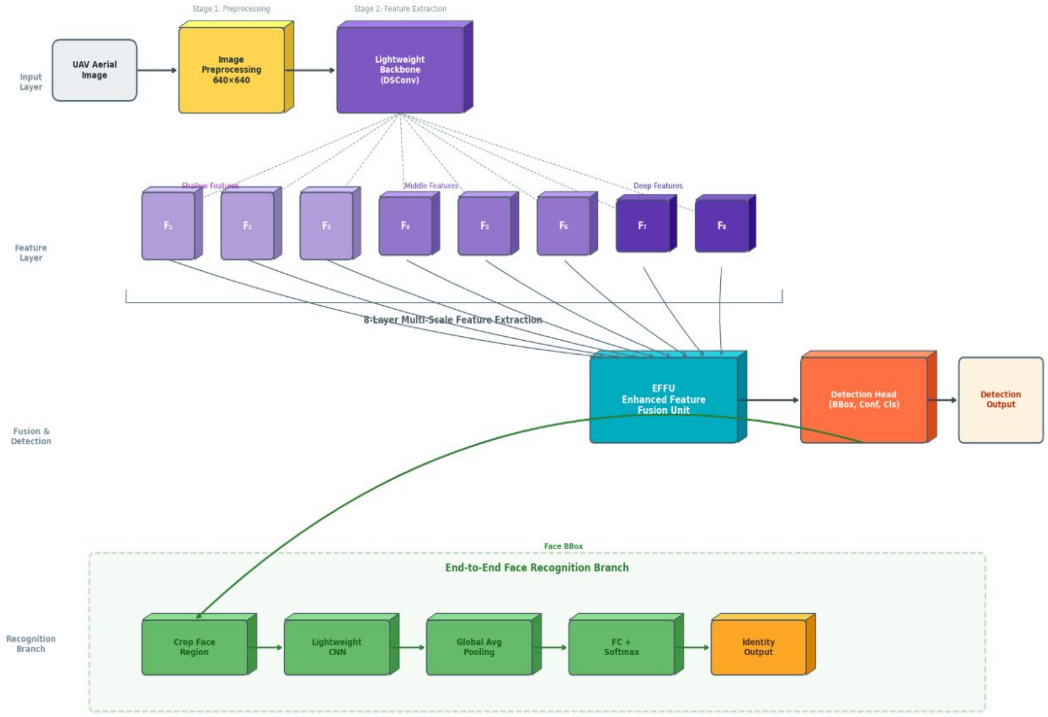


Fig.2 Architecture of the improved face detection network

3) 8-Layer Multi-Scale Feature Extraction

The eight-layer feature extraction network outputs feature maps with channel numbers of 16, 24, 40, 80, 112, 160, 320, and 1280 respectively. Shallow features (F1–F3) have high resolution for small face localization, while deep features (F7–F8) provide strong semantic information for background suppression.

$$\boldsymbol{F} = \{\boldsymbol{F_1}, \boldsymbol{F_2}, \boldsymbol{F_3}, \boldsymbol{F_4}, \boldsymbol{F_5}, \boldsymbol{F_6}, \boldsymbol{F_7}, \boldsymbol{F_8}\} \quad (2)$$

4) Enhanced Feature Fusion Unit (EFFU)

Integrate multi-scale features through up/down-sampling alignment, concatenation, batch normalization, attention weighting, and residual summation, solving scale mismatch and information loss. The Structure of the enhanced feature fusion unit (EFFU) is shown in Fig. 3.

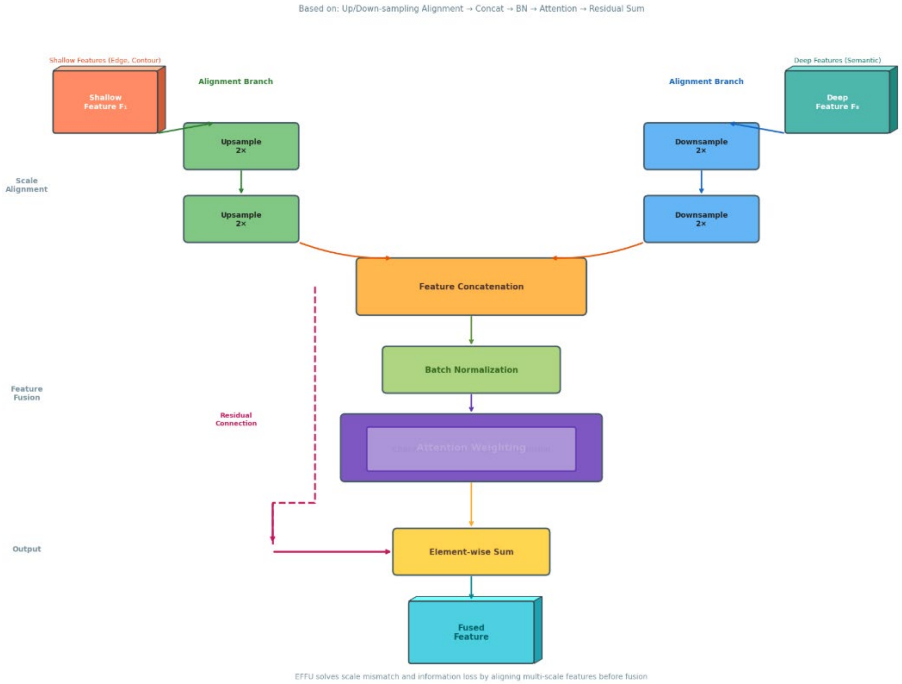


Fig.3 Structure of the enhanced feature fusion unit (EFFU)

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To solve the multi-scale feature mismatch and information loss problem in the original YOLO26, we design an Enhanced Feature Fusion Unit (EFFU). It integrates the 8-layer feature maps through five sequential steps:

Scale Alignment:

$$F_h'=\text{Upsample}(F_h,\text{scale=2,mode=bilinear}) \quad (3)$$

Channel Concatenation:

$$F_{cat}=\text{Concat}([F_h',F_l],\text{dim=1}) \quad (4)$$

Channel Compression & Normalization:

$$F_{conv}=\text{ReLU}(\text{BN}(Conv_{1\times1}(F_{cat}))) \quad (5)$$

Adaptive Attention Weighting:

$$F_{gap}(\text{c})=\frac{1}{H_lW_l}\sum_{i=1}^{H_l}\sum_{i=1}^{W_l}F_{conv}\ (\text{c,i,j}),\text{c=1,2,...,C} \quad (6)$$

$$w=\sigma(FC_2\ (\text{ReLU}(FC_1\ (F_{gap})))) \quad (7)$$

$$F_{att}=F_{conv}\otimes w \quad (8)$$

Residual Connection:

$$F_l'\ =Conv_{1\times1}(F_l) \quad (9)$$

$$F_{effu}=F_{att}+F_l' \quad (10)$$

5) Basic Network Layers and Detection Head

The face detection network consists of convolution, deconvolution, pooling, activation, batch normalization, and residual connection layers, which work together to complete feature extraction, enhancement, and transmission. The detection head includes a full connection layer and a Softmax layer, which are used to predict the face bounding box, confidence, and category.

6) Training

Adopt multi-task loss weighted by CIoU box regression loss, target confidence loss, and classification loss. The mathematical expression of the detection loss is:

$$\mathcal{L}_{det}=\lambda_1\mathcal{L}_{CIoU}+\lambda_2\mathcal{L}_{obj}+\lambda_3\mathcal{L}_{cls} \quad (11)$$

C. Face Recognition

After the detection module outputs face bounding boxes, the corresponding regions are cropped from preprocessed images and fed into this lightweight CNN branch for identity classification. Features are extracted through convolutional layers, compressed via global average pooling, and classified by a Softmax layer to output identity probabilities.

1) Network Architecture

Built on depthwise separable convolution, this lightweight CNN comprises hierarchical convolutional, pooling, batch normalization, activation, global average pooling, and fully connected layers. Tightly coupled with the detection module, it forms an end-to-end framework to avoid error accumulation in separate pipelines and adapt to UAV computing constraints.

2) Feature Extraction

Shallow convolutional layers extract basic facial edges and contours, middle layers capture texture and facial component features (eyes, nose, mouth), and deep layers extract identity-related high-level semantic features. Global average pooling compresses feature maps to reduce parameters while retaining key identity information.

3) Training

The module uses cross-entropy loss for identity classification and adopts a detection-recognition joint training strategy to ensure module synergy. The total loss is the sum of detection and recognition losses.

$$L_{rec}=-\frac{1}{N}\sum_{\{i=1\}}^{N}\sum_{\{j=1\}}^{C}y_j\log(p_j) \quad (12)$$

$$L_{total}=L_{\text{det}+L_{rec}} \quad (13)$$

Where Lrec is recognition loss, N is training samples, C is identity categories, yj is the one-hot label, and pj is the predicted probability.

IV. EXPERIMENTS AND ANALYSIS

To verify the performance of the proposed model in UAV tiny face detection and recognition, controlled experiments are designed, including performance comparison experiments, ablation experiments, and comprehensive analysis, to ensure the fairness and credibility of the experimental results.

A. Experimental Setup

All experiments are conducted on the PyTorch framework with hardware consistent with actual UAV airborne environments. All models use identical input size (640×640), batch size, iteration rounds, and SGD optimizer with cosine annealing learning rate for fair comparison.

B. Evaluation Metrics

Four core metrics are used: Detection Rate (%), Recognition Rate (%), FPS (real-time performance), and Params (M) (lightweight degree).

C. Comparative Experiments

YOLO11-N and the original YOLO26 are selected as comparison models, which are representative one-stage detectors with good real-time performance. All models are trained and tested under the same experimental conditions, and the quantitative results are presented in Table I.

TABLE I. PERFORMANCE COMPARISON ON WIDER FACE DATASET

Model	Backbone	Detection Rate (%)	Recognition Rate (%)	FPS	Params (M)
YOLO11-N	Default	79.2	82.5	41	7.8
Original YOLO26	CSPDarknet53	83.3	87.6	43	8.2
Ours	Lightweight (DSConv-based)	88.6	92.3	35	9.2

Table I shows the performance comparison on the WIDER FACE dataset. The proposed model achieves 88.6% detection rate and 92.3% recognition rate, 5.3% and 4.7% higher than

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original YOLO26, respectively. With only 1.0M additional parameters and 35 FPS real-time performance, it balances accuracy and deployment feasibility.

D. Ablation Experiments

Ablation experiments are designed to verify the effectiveness of each core improved module of the proposed model. On the basis of the original YOLO26 model, the lightweight backbone, 8-layer multi-scale feature extraction, enhanced feature fusion unit (EFFU), and lightweight CNN recognition branch are added step by step, and the performance gain brought by each module is tested. The experimental results are presented in Table II.

TABLE II. ABLATION STUDY RESULTS

Lightweight Backbone	8-layer Feature Extraction	EFFU	CNN Recognition Branch	Detection Rate (%)	Recognition Rate (%)
×	×	×	×	83.3	87.6
✓	×	×	×	85.7	88.9
✓	✓	×	×	86.2	89.5
✓	✓	✓	×	87.5	89.5
✓	✓	✓	✓	88.6	92.3

E. Result Analysis

The proposed model effectively solves the defects of original YOLO26. The lightweight backbone adapts to UAV computing constraints; 8-layer multi-scale features and EFFU suppress background interference and reduce missed/false detections; the end-to-end framework avoids error transmission. The model exhibits strong robustness in complex UAV scenarios.

V. CONCLUSION

This paper proposes a lightweight improved YOLO26 detector to address the performance degradation of face detection and recognition in UAV scenarios caused by tiny faces, motion blur, and complex backgrounds. The model features a DSConv-based backbone, 8-layer multi-scale feature extraction, enhanced feature fusion unit (EFFU), and end-to-end recognition branch. Experimental results on the WIDER FACE dataset show that it achieves 88.6% detection rate and 92.3% recognition rate, outperforming both YOLO11-N and original YOLO26, while maintaining 9.2M parameters and 35 FPS real-time performance. The proposed model balances accuracy and efficiency, making it highly suitable for UAV deployment. Future work will enhance its adaptability to more complex extreme scenarios.

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